1
Introduction

1.1 Quantum chromodynamics

Most high-energy physicists will readily agree that quantum chromodynamics (QCD) is today the well-established theory of strong interactions; at least it has no serious competitor. Quantum chromodynamics is a non-Abelian gauge field theory whose gauge group is the colour group $SU(3)$; it will sometimes be convenient to let the number of colours vary, and to take $SU(N)$ as the gauge group: then the number of colours is N. There are $(N^2 - 1) = 8$ gauge bosons, called gluons, and the matter particles are spin $1/2$ quarks. There are six families of quarks, or six different flavours: up, down, strange, charm, beauty and top. The number of flavours will be denoted by N_f. The last three quarks are heavy and will not play any role at all in the development of this book because their mass is much larger than the characteristic energy scale of a few hundred MeV that we are interested in, while the role of the strange quark will be intermediate. The up and down quarks will often be taken as massless since their mass, of the order of a few MeV, is much smaller than our characteristic energy scale. Note that we use a system of units where $\hbar = c = k_B = 1$, where $\hbar$, c and k_B are, respectively, the Planck constant, the speed of light and the Boltzmann constant. Masses and temperatures will be often measured in MeV or GeV, lengths and times in MeV^{-1} or GeV^{-1}.

Quantum chromodynamics is a renormalizable field theory, and, as in all field theories of this kind, the coupling constant g is a function of the energy scale Q: $g = g(Q)$. The remarkable property of QCD, characteristic of non-Abelian gauge theories, is that the coupling constant tends to zero as the inverse of the logarithm of the energy scale, when this energy scale tends to infinity: this is the famous property of asymptotic freedom, which implies that perturbation theory may be used at high energies, or more precisely in processes with large-momentum transfers, the so-called

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

'hard processes'. This aspect of the theory has been tested in a variety of experiments. The cleanest test is deep inelastic scattering of leptons on nucleons, and there is little doubt that QCD reproduces the pattern of scaling violations observed experimentally. Other processes (lepton pair production, jets, prompt (real) photon production, large-angle scattering, τ-decay,...) have also been studied theoretically and experimentally. No discrepancy with QCD has been found, but the accuracy of the comparison is no better than a few percent, due to theoretical and experimental uncertainties.

When the energy scale decreases, the counterpart of asymptotic freedom is that the coupling constant increases, which leads to a breakdown of perturbation theory as a series expansion in powers of $g^2(Q)$. Thus 'soft' processes, namely, processes with small-momentum transfers, or low-energy properties of hadrons (mass spectrum, resonance widths, scattering lengths,...) cannot be studied in perturbation theory. Actually, this is a matter of principle and not simply a question of the coupling constant being large: assuming massless quarks in order to simplify the discussion, all hadron masses, for example, must be proportional to the QCD scale Λ:

$$\Lambda = \lim_{Q\to\infty} Q\, \mathrm{e}^{-1/2\beta_0 g^2(Q)} [\beta_0 g^2(Q)]^{-\beta_1/2\beta_0^2} \tag{1.1}$$

where g is the (running) gauge coupling constant, β_0 and β_1 are the first two coefficients of the renormalization group β-function: this expression is clearly non-perturbative. Since perturbation theory is irrelevant in low-energy hadron physics, the situation of QCD is much less bright than for hard processes. No good analytic guess has been made, and non-perturbative approaches have been explored following two main avenues:

(i) low-energy models which attempt to incorporate some of the features of the QCD Lagrangian;
(ii) numerical simulations of a discretized version of QCD, that is, lattice QCD.

The most interesting model of type (i) is the effective chiral Lagrangian: it lies on the observation that one of the most conspicuous features of low-energy hadronic physics is the existence of a light particle, that is, the pion. This particle is presumably a Goldstone boson, which would be massless in the case of massless up and down quarks. The corresponding broken symmetry is the chiral $SU(2) \times SU(2)$ symmetry, and the chiral Lagrangian model uses this property as a basic starting point. However, in spite of its usefulness and its phenomenological success, this model, as well as other models, cannot be used in any serious quantitative comparison between QCD and experiment.

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

The other road, that of numerical simulation of lattice QCD, is in principle limited only by computing power. However, in spite of the ingenuity which has been displayed by lattice practitioners in order to keep track of finite-size effects, it will still take some time before anyone is able to reproduce, for example, the hadron spectrum with an accuracy of a few percent. On the other hand, lattice QCD has been decisive in convincing physicists that a phase transition should occur for sufficiently high temperatures and/or densities. At zero temperature, for densities of the order of the nuclear density, quarks and gluons are confined in hadrons: there is no such thing as a free quark or a free gluon propagating in the vacuum. The situation may change when the temperature rises above a critical temperature T_c of the order of 150 MeV; then one may obtain a gas of almost free quarks and gluons, which has been dubbed the quark–gluon plasma. The same situation may occur at zero temperature, for sufficiently large densities (or, more correctly, for sufficiently large chemical potentials), or when one increases both the temperature and the chemical potential. The occurrence of this phase transition is important from a conceptual point of view, as it implies the existence of a new state of matter. In the standard cosmological model, this state of matter prevailed until a time $\sim 10^{-6} - 10^{-5}$ s after the Big-Bang. It could also occur in the core of heavy neutron stars, and it could be produced in collisions of ultrarelativistic heavy ions. Actually, the possibility of producing a quark–gluon plasma in such collisions is the main motivation for interest in the details of this new phase, as one may hope to detect signals of its existence experimentally and to measure some of its properties.

This book will be mainly concerned with the perturbative study of the quark–gluon plasma in an idealized situation where the plasma is at equilibrium, or only slightly out of equilibrium. That a perturbative approach is possible is a consequence of asymptotic freedom: at high temperatures the only scale which is available is the temperature T itself, and the coupling constant $g(T)$ is small at large T: $g(T) \sim 1/\ln(T/\Lambda)$. For large T (very large indeed!), the quark–gluon plasma is close to an ideal gas. However this *a priori* simple situation is spoiled by infrared divergences, which lead to a hierarchy of energy, or length, scales. In addition to the perturbative scale $l_P = 1/T$, there is an electric, or Debye, length scale $l_{\rm el} = 1/gT$, a magnetic length scale $l_{\rm mag} = 1/g^2T$, and perhaps other scales which limit the applicability of naïve perturbation theory.

The same hierarchy of scales is also found in QED plasmas, where the particles are electrons, positrons and ions, and which are relevant, for example, in the core of some stars (see chapter 9). The QCD coupling constant g is replaced by the electron charge e: $l_{\rm el} = 1/eT$, $l_{\rm mag} = 1/e^2T$.

Over the past few years important progress has been made in our understanding of the role of these scales in perturbation theory, and a large

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

part of this book will be devoted to explaining this progress and the present limitations of the theory. Unfortunately, there will be little contact with phenomenology: it is likely that one will have to await the results of the future colliders before a real interaction between theory and experiment is possible. These colliders are the Relativistic Heavy Ion Collider (RHIC) located at Brookhaven, which will provide Au–Au collisions at energies of 200 MeV per nucleon around 1997, and the Large Hadron Collider (LHC), which will be running at CERN at the beginning of the next century and will provide Pb–Pb collisions at energies of a few TeV per nucleon.

1.2 The deconfinement/chiral transition

1.2.1 The bag model

We first examine the possibility of a phase transition in a very simple model, the bag model, which attempts to incorporate, in a crude way, the two main features of QCD: asymptotic freedom and confinement. Hadrons are represented as bubbles of (perturbative) vacuum, or 'bags', in a confining medium. Inside the bag, quarks move freely to a first approximation (in principle one could try to improve the model by taking interactions with gluons in perturbation theory into account, but it is not clear that this strategy is viable, due to the crudeness of the model). Outside the bag, quarks and gluons cannot appear as free particles. This feature is achieved by assuming a constant energy density B, the bag constant, for the vacuum, which keeps quarks and gluons confined. The energy of a hadron is thus composed of the energy associated with the volume of the bag due to the finite energy density of the vacuum, and the kinetic energy of the quarks inside the bag. Assuming, for simplicity, that the bag is spherical with radius R, the kinetic energy is of the form C/R, from the uncertainty principle. Thus the hadron energy E_H is

$$E_H = \frac{4\pi}{3}R^3B + \frac{C}{R} \tag{1.2}$$

By minimizing E_H with respect to R, one finds

$$R = \left(\frac{C}{4\pi B}\right)^{1/4} \tag{1.3}$$

The hadron mass M is given by

$$M = \frac{16\pi}{3}R^3B \tag{1.4}$$

and the pressure P by

$$P = -\frac{\partial E_H}{\partial V} = -B + \frac{C}{4\pi R^4} \tag{1.5}$$

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

The contribution $-B$ due to the vacuum is balanced by the contribution from the quark kinetic energy, so that the total pressure vanishes at equilibrium.

In order to derive the condition for a (deconfinement) phase transition between a hadron phase and a quark–gluon plasma phase, we compare their thermodynamical functions. The grand potential $\Omega = -PV$ in a volume V for an ideal gas of bosons with one degree of freedom is

$$\Omega = VT \int \frac{\mathrm{d}^3k}{(2\pi)^3} \ln\left[1 - \mathrm{e}^{-\beta(\omega_k - \mu)}\right] \tag{1.6}$$

where μ is the chemical potential, V is the volume, $T = \beta^{-1}$ is the absolute temperature and $\omega_k = (k^2 + m^2)^{1/2}$ is the boson energy. For massless bosons and $\mu = 0$, (1.6) yields, after an integration by parts,

$$\Omega = -\frac{VT^4}{6\pi^2} \int_0^\infty \frac{x^3 \mathrm{d}x}{\mathrm{e}^x - 1} = -\frac{\pi^2 V T^4}{90} \tag{1.7}$$

a result that is familiar from blackbody radiation. We write the general result for a gas of massless bosons with zero chemical potential as

$$\Omega = -\nu_b \frac{\pi^2 V T^4}{90} \tag{1.8}$$

where ν_b is the number of degrees of freedom. From equation (1.8), $P = -\partial\Omega/\partial V$ and the relation $E = 3PV$ between the energy and the pressure which is valid for an ultrarelativistic ideal gas, one derives for the pressure and the energy density ε

$$P = \nu_b \frac{\pi^2 T^4}{90} \qquad \varepsilon = \nu_b \frac{\pi^2 T^4}{30} \tag{1.9}$$

Since the chemical potential vanishes by assumption, the entropy S is immediately derived from $\Omega = E - TS$, which leads to an entropy density $s = 4\varepsilon/3T$. We shall also need the particle density deduced from the average number of particles $\overline{N}$ (for one degree of freedom):

$$\overline{N} = V \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\mathrm{e}^{\beta k} - 1} = \frac{VT^3}{2\pi^2} \int_0^\infty \frac{x^2 \mathrm{d}x}{\mathrm{e}^x - 1} = \frac{VT^3}{\pi^2} \zeta(3) \tag{1.10}$$

where $\zeta(p)$ is the Riemann function: $\zeta(3) \simeq 1.202$. Equation (1.10) yields an interesting relation between the particle density $n = \overline{N}/V$ and the entropy density:

$$s = \frac{2\pi^4}{45\zeta(3)} n \simeq 3.6n \tag{1.11}$$

Let us now turn to fermions: the grand potential is

$$\Omega = -VT \int \frac{\mathrm{d}^3p}{(2\pi)^3} \ln\left[1 + \mathrm{e}^{-\beta(\varepsilon_p - \mu)}\right] \tag{1.12}$$

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

with $\varepsilon_p = (p^2 + m^2)^{1/2}$. For massless fermions, taking particles and antiparticles into account leads to the following grand potential, for one degree of freedom:

$$\Omega = -\frac{VT^4}{6\pi^2}\int_0^\infty k^3 \mathrm{d}k \left[\frac{1}{\mathrm{e}^{\beta(\varepsilon_p-\mu)}+1} + \frac{1}{\mathrm{e}^{\beta(\varepsilon_p+\mu)}+1}\right]$$
$$= -\frac{V}{6\pi^2}\left[\frac{7\pi^4 T^4}{60} + \frac{\mu^2\pi^2 T^2}{2} + \frac{\mu^4}{4}\right] \tag{1.13}$$

Let us first examine the possibility of a phase transition with zero chemical potential; in this case the hadronic system will mainly consist of pions. In order to work with simple analytical formulae we neglect the pion mass; the pressure of the hadronic gas is then given by (1.9) with $\nu_b = 3$, corresponding to the three charge states of the pion, while the pressure of the plasma phase is, from (1.9) and (1.13),

$$P = \left(\nu_b + \frac{7}{4}\nu_f\right)\frac{\pi^2 T^4}{90} - B \tag{1.14}$$

In writing (1.14) we have generalized relation (1.5) to a gas of quarks and gluons by adding to the free energy a factor of BV representing the energy of the perturbative vacuum inside the confining medium. The number of bosonic degrees of freedom is $2(N^2-1)$, where N is the number of colours, while $\nu_f = 2NN_f$, N_f being the number of flavours. Inserting the physical values $N = 3$ and $N_f = 2$ corresponding to up and down massless quarks in (1.14) leads to

$$P = \frac{37}{90}\pi^2 T^4 - B \tag{1.15}$$

The equality of the pressure in the plasma and hadron phases gives the critical temperature T_c:

$$T_c = \left(\frac{45B}{17\pi^2}\right)^{1/4} \simeq 144\,\mathrm{MeV} \tag{1.16}$$

with a bag constant $B^{1/4} \simeq 200\,\mathrm{MeV}$ from (1.4). The transition is first-order, and it is a simple exercise to compute the latent heat; note that we use the terminology 'critical temperature' even for first-order phase transitions, although it is often reserved for second-order phase transitions.

In principle the bag model could be used to determine the transition line between the hadronic and plasma phases in the whole (μ, T) plane. Unfortunately, the model is much less convincing for non-zero μ, as is clear

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

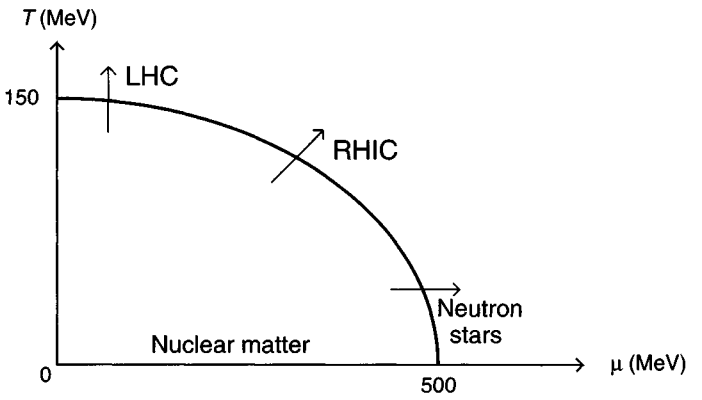


Fig. 1.1 The phase diagram in the μ–T plane.

from considering the $T = 0$ situation. The baryonic density n_B is related to the baryonic chemical potential μ_B and to the nucleon mass m_N through

$$n_B = \frac{2}{3\pi^2}(\mu_B^2 - m_N^2)^{3/2} \tag{1.17}$$

The critical chemical potential can be determined by writing the equality of the pressures together with $\mu_B = 3\mu$, where μ is the quark chemical potential. One finds that the transition is possible only for a narrow range of values of the bag constant (exercise 1.1), which seems difficult to understand physically. Thus the bag model does not seem to give much information on the $T = 0$, $\mu \neq 0$ case. Nevertheless, it is useful to have in mind the following orders of magnitude: for the nuclear density $n_B^{(0)} \simeq 0.15$ fm^{-3}, $\mu \simeq 0.26$ GeV, while for a density equal to five times the nuclear density $\mu \simeq 0.45$ GeV.

We shall see in the next section that lattice QCD confirms the value $T_c \simeq 150$ MeV for the critical temperature with zero chemical potential. Unfortunately, there is no reliable estimate of the critical chemical potential at zero temperature. It is generally assumed that the transition should occur for a density of three to five times the nuclear density, so that we may guess the phase diagram of fig. 1.1 in the (μ, T) plane; the regions in this plane which are reached in various physical situations are also indicated on the plot.

1.2.2 *Lattice quantum chromodynamics*

The discretized version of QCD can be studied on computers using Monte-Carlo simulations. The details of 'how one puts QCD on a lattice' will not be given here, as a detailed account of lattice QCD is outside the scope

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

of this book; the interested reader is referred to the literature: see, for example, Creutz (1983) or Karsch (1990, 1992). Simulations with dynamical quarks are much more time consuming than those without quarks, and this explains why many computations have been performed in the pure gauge sector by taking only the gluon degrees of freedom into account.

Phase transitions are often associated with broken symmetries, and, of course, symmetries of the lattice action play a crucial role. The action S_G of a pure gauge theory with a gauge group $SU(N)$ has a global $Z(N)$ symmetry, where $Z(N)$ (the Abelian group of the Nth-roots of unity) is the centre of the group, namely, the subgroup whose elements commute with all group elements. This symmetry is spontaneously broken at the phase transition and the corresponding order parameter is the expectation value of the Polyakov loop L (see, for example, Karsch (1990) for a definition). This expectation value $\langle L \rangle$ is related to the excess of free energy $F_q(T)$ induced by a static quark source in the gluonic heat bath:

$$\exp(-F_q(T)/T) \simeq \langle L \rangle \tag{1.18}$$

In the confined phase $F_q(T)$ is infinite since a coloured quark cannot be screened, and $\langle L \rangle = 0$. On the other hand, in the deconfined case $F_q(T)$ is finite and $\langle L \rangle \neq 0$. Note that the order parameter is zero in the low-temperature phase and non-zero in the high-temperature phase.

When quarks are added to the Lagrangian, global $Z(N)$ symmetry is lost. However, in the case of massless quarks, there is another global symmetry of the action, that is, chiral symmetry, with symmetry group $SU(N_f) \times SU(N_f)$, which is spontaneously broken at low temperature. The (local) order parameter in this case is the quark condensate $\langle \overline{\psi}\psi \rangle$, where ψ is the quark field. The quark condensate is non-zero at low temperature, where chiral symmetry is broken, and zero above the phase transition temperature, where chiral symmetry is restored.

With massive quarks, no global symmetry of the action survives. Nevertheless, numerical simulations indicate a sharp rise in entropy, energy density and pressure around a temperature T_c, which is thought to be the critical temperature for the deconfinement/chiral symmetry restoration phase transition. There is no indication of two different transitions, one corresponding to deconfinement and the other to chiral symmetry restoration, at least in the case of zero chemical potential. Let us give a brief summary of what has been learned in recent years on the following topics:

(i) the nature of the phase transition;
(ii) the value of the critical temperature;
(iii) the equation of state above the critical temperature;
(iv) the Debye screening length.

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

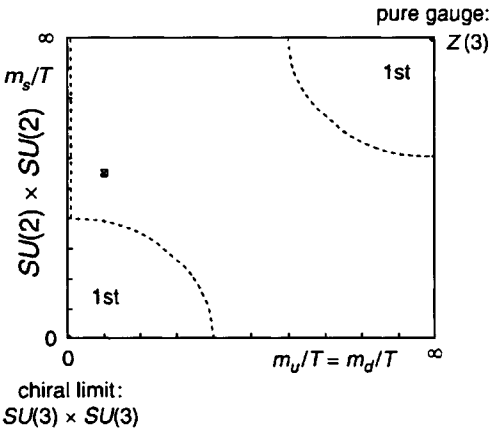


Fig. 1.2. Character of the transition as a function of the light quark masses $m_u = m_d$ and the strange quark mass m_s. The circle with a cross represents the physical values of the parameters. Private communication from F. Karsch.

In the case of a pure gauge theory, there are convincing universality arguments which allow one to relate the $SU(N)$ deconfinement transition in dimension four to the ferromagnetic–paramagnetic transition of a $Z(N)$ spin model with ferromagnetic interactions in dimension three. In the $SU(2)$ case, the spin model is the Ising model and the phase transition is well-known to be second order. There is remarkable agreement between the numerical values of the critical exponents of the $SU(2)$ deconfinement transition and those of the three-dimensional Ising model, which gives a beautiful confirmation of universality. In the $SU(3)$ case, the spin model is the three-state Potts model, whose transition is first order. A careful analysis of the deconfinement transition, using finite-size scaling, does confirm the first-order character of the deconfinement transition.

In the case of chiral symmetry restoration, it is also possible to use universality arguments in order to study the character of the transition, by relating the gauge theory to simpler models. The transition can be shown to be first order for $N_f \geq 3$, while for $N_f = 2$ the transition is probably second order, although the situation is not entirely clear.

Our present knowledge of the character of the transition, which relies on the preceding universality arguments and on numerical simulations, is summarized in fig. 1.2. At present, with two massless up and down quarks and a strange quark with a mass around 150 MeV, the best guess seems to be that the deconfinement/chiral transition is second order. However, one must remember that, without symmetry breaking, it is always possible that the transition between the hadronic and plasma phases occurs gradually and a critical temperature may not be precisely defined.

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

Let us now turn to the critical temperature. With present numerical simulations, asymptotic scaling is still not established for $SU(2)$ and $SU(3)$ gauge theories; however scaling violations drop out in the ratio of physical quantities, so that one can compute $T_c/\Lambda_{\overline{MS}}$, or $T_c/\sqrt{\sigma}$, or T_c/m_ρ, where $\Lambda_{\overline{MS}}$, σ and m_ρ are the QCD mass scale (see (1.1)) in the so-called $\overline{MS}$ renormalization scheme, the string tension and the ρ-meson mass respectively. It seems that the best estimates are deduced from the T_c/m_ρ ratio and one finds, in MeV,

$$\begin{aligned} &-\text{pure gauge } SU(3): \quad T_c = 240 \pm 20 \text{ MeV} \\ &-N_f = 2 \qquad\qquad\quad : \quad T_c = 150 \pm 10 \text{ MeV} \end{aligned}$$

In numerical simulations of the equation of state, it is very important to proceed in a fully non-perturbative manner. Up to now, this has been possible only in the pure gauge sector. From the expectation value of the plaquette action and the non-perturbative determination of the renormalization-group β-function, one computes the pressure and the quantity $\Delta = (\varepsilon - 3P)/T^4$ which measures the deviation with respect to the ideal gas behaviour $\varepsilon = 3P$. The results are first obtained as functions of the coupling $2N/g^2$, which may then be converted into T/T_c thanks to the non-perturbative β-function. Figure 1.3 shows the behaviour of the ratios ε/T^4 and $3P/T^4$, compared with the value $\varepsilon_{SB}/T^4 = 8\pi^2/15$ of an ideal gluon gas, and the value $\simeq 6.2$ on an $N_\tau = 6$ lattice. Even at $T = 2T_c$, the energy density only reaches about 70% of the ideal gas value. However, at least for $T \gtrsim 2T_c$, one finds $\varepsilon \simeq 3P$.

Let us finally examine screening lengths. In QCD a naïve perturbative calculation with resummed self-energy bubbles for the gluon propagator yields the following behaviour of the potential $V(r)$ between gauge invariant sources:

$$V(r) \propto \frac{\mathrm{e}^{-2m_{\mathrm{D}}r}}{r^2} \tag{1.19}$$

where the Debye mass $m_{\mathrm{D}} = l_{\mathrm{el}}^{-1}$ is given by (see chapter 6)

$$m_{\mathrm{D}}^2 \simeq \frac{1}{3}\left(N + \frac{1}{2}N_f\right) g^2(T) T^2 \tag{1.20}$$

The behaviour (1.19) is controlled by $2m_{\mathrm{D}}$ because at least two gluons must be exchanged between two gauge-invariant sources. The Debye mass can be estimated in numerical simulations by computing the large-distance behaviour of the Polyakov loop correlation function. For reasons of computing time, this can be done in practice only in the case of pure gauge theories. One finds for $T \geq T_c$, both in $SU(2)$ and $SU(3)$, that $m_{\mathrm{D}} \simeq T$ within a 10% accuracy, which confirms the proportionality with

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

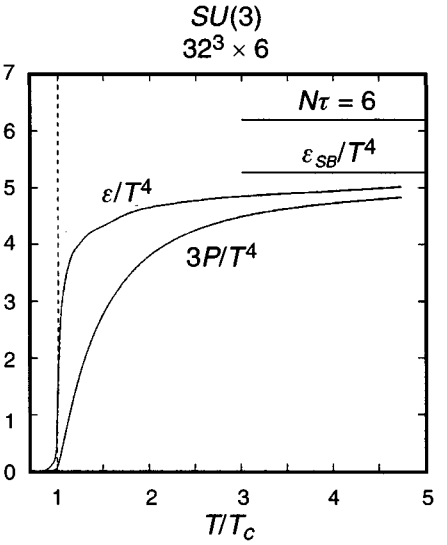


Fig. 1.3. Ratios ε/T^4 and $3P/T^4$ as functions of T/T_c on the lattice. Private communication from F. Karsch.

T of equation (1.20) and is even in quantitative agreement with (1.20) if one chooses $g(T) \simeq 1$.

1.3 Heavy ion collisions

As has already been mentioned, the main reason for being interested in 'hot QCD' is that one may hope to produce a quark–gluon plasma in collisions of ultrarelativistic heavy ions. In this section, we wish to show that such collisions are indeed able to produce energy densities that are sufficiently large for the phase transition to occur. In order to estimate the available energy density, we shall give a short account of the space-time and thermal evolution of the collision. In what follows, we always view the collision in the centre of mass frame; E_{cm} denotes the energy per nucleon in this frame, the z-axis is chosen along the collision axis and we neglect transverse motion. Finally, for simplicity, we restrict ourselves to collisions of identical nuclei with atomic number A.

The generally accepted picture for the collision of two ultrarelativistic heavy nuclei rests on two concepts which are consistent with the present experimental data: nuclear transparency and the inside–outside cascade.

Nuclear transparency, or the leading particle effect, was first observed in nucleon–nucleon collisions: a large fraction (about a half) of the available energy is carried away by two secondary nucleons, which are called the leading baryons. Similarly, in the collision of two heavy nuclei, the baryonic number is carried away essentially by two systems of nucleons

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

leaving the interaction region with large momenta in the centre of mass frame. More quantitatively, the maximum rapidity in the collision is $Y = \ln(2E_{\rm cm}/m_N)$; the rapidity distribution of the baryonic number is centred at $\pm(Y - \delta y)$. Nucleon–nucleon collisions give $\delta y \simeq 2$ units of rapidity, but event generators suggest a somewhat larger value, of the order of 3–3.5, for A–A collisions. There is a range $\simeq [-(Y - 2\delta y), (Y - 2\delta y)]$ where the baryonic number averages to zero. This is the so-called central region, while the regions in rapidity which carry the baryonic number are called the fragmentation regions. For a clear separation of central and fragmentation regions, one probably needs at least $Y \simeq 5 - 6$, namely, $E_{\rm cm} \geq 100$ GeV.

The concept of inside–outside cascade rests on the idea that an excited hadronic system does not decay instantaneously: in its rest frame, the characteristic decay time is denoted by τ_0, with $\tau_0 \simeq 1$ fm/c. Because of time dilation, slow systems disintegrate earlier than faster ones: this is the inside–outside cascade. Assume that an excited hadronic system leaves the interaction point $z = 0$ at time $t = 0$ and moves freely with velocity v along the collision axis. Its proper time is $\tau = (t^2 - z^2)^{1/2} = t(1 - v^2)^{1/2}$. A collection of hadronic systems created at $t = z = 0$ will thus disintegrate on average on a hyperbola of constant proper time $(t^2 - z^2)^{1/2} = \tau_0$; in a hydrodynamical description, the initial conditions for the fluid will be given on this hyperbola. Note that the initial conditions are not given at an initial time, but, rather, at an initial proper time.

We may now summarize our picture of the collision: two Lorentz contracted nuclei, with transverse radius R and longitudinal extension $d_L \sim 2Rm_N/E_{\rm cm}$ collide during a time $\sim d_L/c$. The two nuclei cross each other and two Lorentz contracted hadronic systems, carrying the initial baryonic number, recede from each other at high velocity in opposite directions from the interaction region. They leave behind them excited hadronic matter with zero baryonic number. It is interesting to observe that for Pb–Pb collisions at LHC energies, with $E_{\rm cm} = 2$ TeV, $d_L/c \sim 7 \times 10^{-3}$ fm/c $\ll \tau_0$, so that the duration of the collision may be neglected. For lower energies, it may be necessary to take the finite time of crossing into account.

We shall now limit ourselves to the study of the central hadronic system, which carries zero baryon number. Recall that in our picture a collection of excited hadronic systems leaves the origin at $t = 0$ and disintegrates on a hyperbola $(t^2 - z^2)^{1/2} = \tau_0$; the coordinate which parametrizes a point on this hyperbola is denoted by α:

$$\alpha = \frac{1}{2} \ln \frac{t + z}{t - z} \tag{1.21}$$

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

and it can be identified with the rapidity y, since our hadronic systems are assumed to move freely. Experimental data indicate that the particle density is approximately independent of the rapidity in the central region. We may thus assume that the initial conditions on the hyperbola $\tau = \tau_0$ are invariant under Lorentz boosts of moderate velocity (in order not to leave the central region), that is, they do not depend on y: these are Bjorken's initial conditions.

In order to predict the subsequent evolution of the collision, we assume the validity of a hydrodynamical picture, with a relativistic perfect fluid; these assumptions will be discussed at the end of the section. Note that, for the time being, we do not specify the character of the fluid, which may be hadronic-like or plasma-like. The rest frame of the fluid is defined as the frame in which the energy flux is zero. We denote the four-velocity by u^μ, $u^\mu = (1 - v^2)^{-1/2}(1, \mathbf{v})$, where $\mathbf{v}$ is the fluid velocity in the centre of mass frame of the collision. Let us consider a locally conserved scalar density N, and its volume density n measured in the rest frame of the fluid. In the case of a perfect fluid, there are no transport phenomena, so that there is no energy flux without scalar flux. Thus, in the rest frame of the fluid, the flux of any conserved quantity is zero. This leads to the local conservation law:

$$\partial_\mu(nu^\mu) = 0 \tag{1.22}$$

Similarly, we write the conservation of the energy-momentum tensor $T^{\mu\nu}$ as

$$\partial_\mu T^{\mu\nu} = 0 \tag{1.23}$$

In the fluid rest frame, we have $T^{00} = \varepsilon$, $T^{ij} = P\delta_{ij}$, where, as before, ε is the energy density and P is the pressure. From Lorentz covariance one deduces the well-known expression that is valid in any frame:

$$T^{\mu\nu} = (\varepsilon + P)u^\mu u^\nu - Pg^{\mu\nu} \tag{1.24}$$

Equations (1.22)–(1.24), together with the equation of state, form a closed system. Let us now derive from this formalism an equation for entropy conservation. Contracting (1.23) with u^ν yields

$$u^\mu(\partial_\mu\varepsilon) + (\varepsilon + P)\partial_\mu u^\mu = 0 \tag{1.25}$$

On the other hand, if there are conserved scalar quantities N_i, with associated densities n_i and chemical potentials μ_i, we have thermodynamical identities:

$$\mathrm{d}\varepsilon = T\mathrm{d}s + \sum_i \mu_i \mathrm{d}n_i \tag{1.26a}$$

$$\varepsilon + P = Ts + \sum_i \mu_i n_i \tag{1.26b}$$

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

where s is the entropy density. Using (1.22) and (1.26) and assuming all densities to be continuous functions of space-time allows us to derive the equation for entropy conservation:

$$\partial_\mu(su^\mu) = 0 \tag{1.27}$$

We are now in a position to describe the longitudinal expansion of the hadronic matter produced in the central region, ignoring transverse expansion and making use of Bjorken's initial conditions. From the Lorentz invariance of our hydrodynamical equations and of our initial conditions, the properties of the fluid in a slice around the longitudinal coordinate z can be deduced from those of the fluid in a slice around $z = 0$ by a longitudinal Lorentz boost. It is convenient to use the coordinates (τ, α), with $t = \tau \cosh \alpha$, $z = \tau \sinh \alpha$. In this coordinate system, the components (V^τ, V^α) of a four vector V^μ read

$$\begin{aligned} V^\tau &= V^0 \cosh \alpha - V^z \sinh \alpha \\ V^\alpha &= -V^0 \sinh \alpha + V^z \cosh \alpha \end{aligned} \tag{1.28}$$

from which follows

$$\partial_\mu V^\mu = \frac{1}{\tau}\left(\frac{\partial(\tau V^\tau)}{\partial \tau} + \frac{\partial V^\alpha}{\partial \alpha}\right) \tag{1.29}$$

The invariance of the initial conditions under Lorentz boosts, together with the Lorentz invariance of the hydrodynamical equations, requires that no quantity depends on α: see exercise 1.2 for further details. A particular solution of the hydrodynamical equations is given by $u^\alpha = 0$, $u^\tau = 1$; in this case the fluid rapidity y coincides with the space-time rapidity α while equation (1.27) for entropy conservation becomes

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(\tau s) = 0 \tag{1.30}$$

This implies that the entropy density decreases as the inverse of the proper time τ. Since the system is expanding in the z-direction, the volume of a slab of fluid in the rapidity range $[y, y + \mathrm{d}y]$ grows as τ : $\mathrm{d}^3x = \tau \mathrm{d}y\, \mathrm{d}^2x_\perp$ in a frame where the fluid is at rest. Equation (1.30) means that the entropy contained in a given rapidity interval is conserved.

If we assume that the system thermalizes at a proper time $\tau_0 \simeq 1$ fm/c, with an initial entropy density s_0, we shall have for $\tau \geq \tau_0$

$$s(\tau) = \frac{s_0 \tau_0}{\tau} \tag{1.31}$$

We now turn to numerical estimates. The total entropy contained in a slice $[y, y + \Delta y]$ at a proper time τ_0 is

$$\Delta S = \pi R^2 s_0 \tau_0 \Delta y \tag{1.32}$$

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

where the nuclear radius $R \simeq 1.2 \times A^{1/3}$. Equation (1.11) allows one to relate s_0, or more properly $s_0\tau_0$, to an observable, namely, the rapidity density $\mathrm{d}N/\mathrm{d}y$: as the baryonic number vanishes in the central region, entropy is mainly carried by pions and we have

$$s_0\tau_0 \simeq \frac{3.6}{\pi R^2}\frac{\mathrm{d}N}{\mathrm{d}y} \tag{1.33}$$

The rapidity density $\mathrm{d}N/\mathrm{d}y$ in A–A collisions can be estimated from

$$\frac{\mathrm{d}N}{\mathrm{d}y} \simeq A^{1.1}\frac{\mathrm{d}N_0}{\mathrm{d}y} \qquad \frac{\mathrm{d}N_0}{\mathrm{d}y} \simeq 0.9 \ln \frac{E_{\mathrm{cm}}}{m_N} \tag{1.34}$$

where $\mathrm{d}N_0/\mathrm{d}y$ is the rapidity density in nucleon–nucleon collisions. The exponent of A in (1.34) reflects multiple scattering, and could be slightly higher than the conservative estimate of 1.1. With $\tau_0 \simeq 1$ fm/c, we obtain for Pb–Pb collisions

$$s_0 \simeq 7.5 \ln \frac{E_{\mathrm{cm}}}{m_N}\ \mathrm{fm}^{-3} \tag{1.35}$$

For $E_{\mathrm{cm}} = 2$ TeV, in the case of an ideal quark–gluon plasma with $N_f = 2$, this value corresponds to a temperature $T_0 \simeq 300$ MeV, about twice the critical temperature, and to an energy density $\varepsilon \simeq 12.8\,\mathrm{GeV/fm}^3$.

It is also interesting to follow the evolution of the temperature; in the case of zero chemical potential we have from (1.26)

$$\frac{s}{T}\frac{\partial T}{\partial s} = \frac{\partial P}{\partial \varepsilon} = c_s^2\left(= \frac{1}{3}\right) \tag{1.36}$$

where c_s is the velocity of sound ($c_s = 1/\sqrt{3}$ for an ideal ultra-relativistic gas). From entropy conservation (1.30) we obtain

$$T = T_0\left(\frac{\tau_0}{\tau}\right)^{c_s^2} \tag{1.37}$$

which shows that the temperature drops more slowly than the entropy density.

From the preceding discussion we may now give the following picture of the central region. At a proper time $\tau_0 \simeq 1$ fm/c after the collision, thermal equilibrium is reached and the temperature is large enough for the system to be in the quark–gluon plasma phase. The assumption of thermal equilibrium is difficult to justify in an entirely convincing way, but it is not unreasonable: for a quark–gluon plasma at $T \simeq 200$ MeV, one has approximately a particle density n of four particles per fm^3; if the cross-section σ is estimated from the additive quark model to be around 10 mb, the mean free path $\lambda = 1/n\sigma \sim 0.25$ fm, so that the size of the interaction region is 15–20 times λ for Pb–Pb collisions. However, a recent discussion suggests that the situation could be more complicated:

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

the gluons might reach equilibrium much faster than the quarks, because the gluon–gluon cross-section, as computed in perturbative QCD, is much larger than the gluon–quark and quark–quark cross-sections. The gluon mean free path is estimated to be around 0.3 fm, while the quark mean free path is around 2 fm. If only the gluons come to thermal equilibrium, the initial temperature is estimated to be around 400 MeV, instead of the 300 MeV found earlier.

Thermalization justifies the subsequent use of hydrodynamics, and the evolution is obtained by assuming the perfect fluid approximation, where entropy is conserved since there is no dissipation. The system expands mainly in the longitudinal direction, and the temperature drops until it reaches T_c. Then a mixed plasma–hadronic phase occurs which lasts for some time until a pure hadronic phase is obtained. Finally, when the mean free path is of the order of the size of the system, free particles are produced and may be observed in detectors; this is the so-called freeze-out transition. The assumptions which underly this hydrodynamical description seem less crucial than that of thermalization. Entropy may be produced by viscous effects, shock waves which develop from the phase transition and the freeze-out transition. This entropy production has been estimated to be small and, in any case, the main virtue of the hydrodynamical approach is that it includes the fundamental conservation laws: most of the results can be viewed as consequences of the basic conservation laws, and one probably need not assume the detailed validity of perfect fluid dynamics.

References and further reading

Standard references to statistical mechanics are Huang (1962), Reif (1965), Ma (1985) or Mandl (1988). A good introduction to QCD may be found in the book by Muta (1987) and recent results are reviewed in the articles collected in the proceedings of the conference *QCD: 20 Years Later*; see in particular the contributions by Wilczek (1992) and Altarelli (1992). Lattice QCD is explained in the book by Creutz (1983). More recent reviews, with emphasis on finite temperature, are Karsch (1990, 1992). The universality arguments relating lattice gauge theory to $Z(N)$ models are given by Svetitsky and Yaffe (1982), while analogous arguments in the case of chiral transition are due to Pisarski and Wilczek (1984). The general space-time picture of heavy ion collisions was first developed by Bjorken (1983). See also the review article by Cleymans, Gavai and Suhonen (1986), the book by Shuryak (1988) and the articles by Satz (1990, 1992). A very clear account of the hydrodynamical picture is to be found in Blaizot and Ollitrault (1990); see also the original article by Baym, Friman, Blaizot, Soyeur and Czysz (1983). The discussion of equilibration times is due to Shuryak (1992).

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press

Exercises

1.1 Discuss the transition between the hadronic phase and the plasma phase at zero temperature in the framework of the bag model. First compute the pressure and the energy density of the hadronic phase as functions of the chemical potential μ and the nucleon mass m_N. Show that the transition is possible only if

$$\frac{1}{2\pi^2}\left(\frac{m_N}{3}\right)^4 \leq B \leq \left(\frac{3}{4} - \ln 2\right)\frac{m_N^4}{8\pi^2}$$

where B is the bag constant.

1.2 Assuming that the fluid four-velocity may be written as $u^\mu = (\cosh\theta, \sinh\theta)$, where we neglect transverse dependence and where θ is *a priori* a function of (t, z), show that the hydrodynamical equations read

$$\tau\frac{\partial\varepsilon}{\partial\tau} + \tanh(\theta - y)\frac{\partial\varepsilon}{\partial y} + (\varepsilon + P)\left(\frac{\partial\theta}{\partial y} + \tanh(\theta - y)\tau\frac{\partial\theta}{\partial\tau}\right) = 0$$

$$\frac{\partial P}{\partial y} + \tanh(\theta - y)\tau\frac{\partial P}{\partial\tau} + (\varepsilon + P)\left(\tau\frac{\partial\theta}{\partial\tau} + \tanh(\theta - y)\frac{\partial\theta}{\partial y}\right) = 0$$

with $t = \tau\cosh y$, $z = \tau\sinh y$. Also write the equation for baryon number conservation. Show that the above equations lead to (1.30) when one uses Bjorken's initial conditions $\theta(y, \tau_0) = y$.

https://doi.org/10.1017/CBO9780511721700.002 Published online by Cambridge University Press